

Question 1 (1 point) ✓ Saved

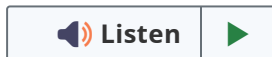


What is the main objective of linear regression?

The main objective of linear regression is to predict a numerical value based off of other variables.



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How does Ridge Regression differ from standard linear regression?

Ridge Regression introduces a penalty term to the cost function, which forces the model to use smaller weights which allows it to adapt better to new data.



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What is the impact of the α parameter in Ridge and Lasso Regression?

It allows you to regularize the model.



Question 4 (1 point) ✓ *Saved*



Why might Lasso Regression be preferred over Ridge Regression in some cases?

Ridge Regression never makes its weights zero, while Lasso Regression does, meaning Lasso Regression can eliminate unnecessary variables.



Question 5 (1 point) ✓ *Saved*



What is the purpose of the learning rate in gradient descent?

The learning rate determines the size of the gradient descent algorithm steps when minimizing the cost function. Its value can determine how fast/slow the algorithm will find the solution, although it being too extreme can cause problems.



Question 6 (1 point) ✓ Saved



Explain the difference between Batch and Stochastic Gradient Descent.

The Batch Gradient Descent uses the entire dataset at each step while the Stochastic Gradient Descent only uses a single instance.



Question 7 (1 point) ✓ Saved



What is overfitting, and how does regularization help?

Overfitting is when a model learns the exact values of the dataset and does not generalize to new data. Regularization constrains the model so it can't become overfit to the data.

A ✓

Question 8 (1 point) ✓ *Saved*

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How does Lasso Regression help in feature selection?

Lasso Regression can make some of the weights 0, meaning those features are not calculated into the final calculation, which is effectively feature selection.

A ✓

Question 9 (1 point) ✓ *Saved*

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When should you prefer Ridge Regression over Lasso Regression?

Ridge Regression can converge faster than Lasso Regression.



Question 10 (1 point) ✓ *Saved*

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What is the formula for the cost function in Ridge Regression?

$J(\theta) = \text{MSE}(\theta) + \alpha/m * \sum_{i=1}^n (\theta_i)^2$



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